This is an illustration of inverse variation or indirect variation. The graph is a part of a curve called Rectangular Hyperbola.

Example 3.49 A company initially started with 40 workers to complete the work by 150 days. Later, it decided to fasten up the work increasing the number of workers as shown below.

Number of workers (x)	40	50	60	75
Number of days (y)	150	120	100	80

(i) Graph the above data and identify the type of variation.

(ii) From the graph, find the number of days required to complete the work if the company decides to opt for 120 workers?

(iii) If the work has to be completed by 200 days, how many workers are required?

(i)

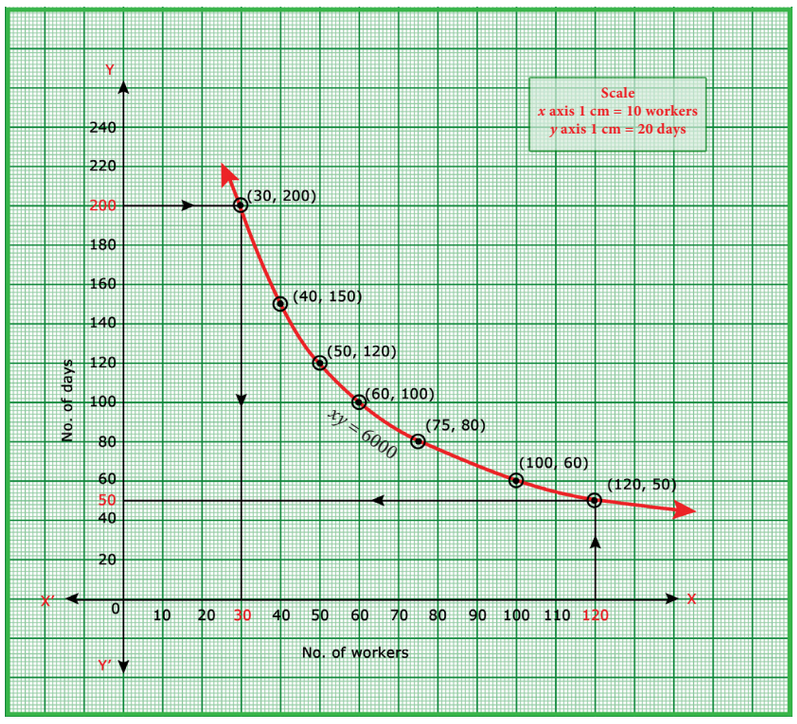


Fig. 3.13

From the given table, we observe that as x increases, y decreases. Thus, the variation is an inverse variation.

Let $y = \frac{k}{x}$

$\Rightarrow xy = k$, $k > 0$ is called the constant of variation.

From the table, $k = 40 \times 150 = 50 \times 120 = \ldots = 75 \times 80 = 6000$

Therefore, $xy = 6000$

Plot the points (40,150), (50,120), (60,100) of (75,80) and join to get a free hand smooth curve (Rectangular Hyperbola).

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